

Monolithic beam / rigid-body coupling in beamFoam and moorFV

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beamFoam `colm_6DOF_solver` → `openfoam-v2306`, PR #49 (draft) **moorFV** `feature-beamFoamCoupling-colm` → `main`, PR #16 (draft)

Goal. When a rigid body is attached to the end of a beam, solve the body's motion inside beamFoam's own linear system (monolithically), instead of taking turns between beamFoam and the `sixDoFRigidBodyMotion` solver.

Status. beamFoam now solves the body's 6 DOFs in the same Eigen system as the beam, and that solution drives the body in moorFV. The stable mode (`staged`) reproduces the old partitioned result at the same cost on the single-beam benchmark. The coupling is not yet fully implicit: the beam-end position lags by one Newton iteration, and torque is still added explicitly.

1. Old scheme and new scheme

Before: partitioned

1. `sixDoFRigidBodyMotionFvBeam` updates the body.
2. The `finiteVolumeBeam` restraint prescribes the beam-end displacement `W` from the body.
3. `beam.evolve()` solves the beam.
4. The attachment force is returned to the 6DOF solver as a restraint force.

The two solvers exchange data once per outer iteration.

Now: monolithic (switch-gated)

1. `finiteVolumeBeam` sends the full body state to beamFoam (`RigidBodyStepData`).
2. `BlockEigenSolver0F` solves the beam unknowns *and* 6 body unknowns in one sparse system.
3. moorFV reads the solved body state and applies it to the body (`applyBlockEigenSolution`).
4. The restraint returns zero force, so the beam force is not applied twice.

Matrix structure

The beam system (unknowns `W`, `θ` per cell) gets six extra unknowns: body displacement and rotation corrections. The body rows use the same Newmark- β update as the old 6DOF solver, so the body's diagonal block is the identity and the Newmark terms go on the right-hand side.

$$\begin{bmatrix} K_{\text{beam}} & K_{\text{beam} \rightarrow \text{rb}} \\ K_{\text{rb} \rightarrow \text{beam}} & I_{6 \times 6} \end{bmatrix} \begin{bmatrix} \Delta W, \Delta \theta \text{ (beam)} \\ \Delta x_{\text{rb}}, \Delta \theta_{\text{rb}} \text{ (body)} \end{bmatrix} = \begin{bmatrix} \text{beam RHS} \\ \text{Newmark body RHS} \end{bmatrix}$$

$$x(n+1) = x(n) + v(n) \cdot \Delta t + \Delta t^2 \cdot [\beta \cdot a(n+1) + (\tfrac{1}{2} - \beta) \cdot a(n)]$$

- **Beam** → **body** (`K_beam→rb`): beam attachment rows depend on body translation (`Cw/pDelta`) and on body rotation through the moment arm (`(Cw · -spin(r))/pDelta`).
- **Body** → **beam** (`K_rb→beam`, the "Hooke's law" coefficient, 11 Aug): body translation rows get $\alpha K \cdot (\Delta x_{\text{rb}} + \Delta \theta_{\text{rb}} \times r - \Delta W_{\text{beamCell}})$, where $\alpha K = \Delta t^2 \cdot \beta \cdot (1/m) \cdot CQW/L$, i.e. roughly EA/L scaled by the

Newmark factor. $L = p\Delta t$ is the face-to-cell-centre distance at the attachment. This replaced an earlier version that left out the $\Delta t^2 \beta / m$ scaling and was dimensionally inconsistent.

- **Torque** is not in the matrix; it is added explicitly through `currTorque`. An implicit version would need a coefficient based on the moment of inertia instead of $1/m$.

2. What changed in each repository

beamFoam (PR #49)

File	Change
<code>BlockEigenSolverOF.C/.H</code>	Adds the 6 rigid-body rows and columns, Newmark body RHS, beam↔body coupling blocks, and a full <code>RigidBodySolution</code> output (displacement, rotation, v , a , π , τ). Writes <code>BlockEigenSolve_rigidBodyMotion.dat</code> and <code>ForceCoupling.dat</code> .
<code>RigidBodyCouplingData.H</code> (new)	<code>RigidBodyStepData</code> (body state in) and <code>RigidBodySolution</code> (body state out).
<code>beamModel.H</code>	New virtuals <code>setRigidBodyStepData()</code> and <code>getRigidBodySolution()</code> , so <code>moorFV</code> doesn't need to know the concrete beam type.
<code>coupledTotalLagNewtonRaphsonBeam*</code>	Reads the coupling switches, prepares coupling coefficients, applies the staged attachment target before matrix assembly, adds configuration checks (errors/warnings).
<code>original_BlockEigenSolverOF.C/.H</code>	Backup of the pre-change solver; not compiled.
<code>tutorials/rigidBodyAndBeam*</code>	Seven test and benchmark cases, plus comparison, regression and efficiency plotting scripts.

moorFV (PR #16)

File	Change
<code>finiteVolumeBeam.C/.H</code>	Builds <code>RigidBodyStepData</code> from <code>state()</code> / <code>state0()</code> and passes it to the beam; <code>beamAttachmentKinematics</code> option; applies the BlockEigen solution and zeroes the restraint force when authoritative; new <code>attachmentForce</code> and <code>displacement</code> history files.
<code>sixDoFRigidBodyMotionFvBeam.C/.H/I.H</code>	New <code>applyBlockEigenSolution()</code> (sets centre of rotation, v , a , π , τ , Q , with constraints applied); new <code>state0()</code> accessor; <code>initialCentreOfRotation()</code> made public.
<code>sixDoFRigidBodyMotionFvBeamRestraint.H</code>	Interface change: <code>restrain(...)</code> now takes a non-const motion object so a restraint can update the body.
<code>Make/options</code> (x2)	Include beamFoam headers through <code>\$(BEAMFOAM_DIR)</code> . Hard-coded <code>/Users/...</code> path removed on 30 Sep (74005ba).

3. Configuration switches

Switch (file)	Values (default first)	Effect
<code>blockEigenForceCoupling</code> <code>beamProperties</code>	<code>false</code> / <code>true</code>	Includes the beam attachment force in the body rows.
<code>blockEigenKinematicCoupling</code> <code>beamProperties</code>	<code>false</code> / <code>true</code>	Adds the beam↔body kinematic coupling blocks to the matrix.
<code>blockEigenKinematicCouplingMode</code> <code>beamProperties</code>	<code>staged</code> / <code>fullyImplicit</code>	Staged: attachment target fixed at the last accepted body state during the Newton loop. Fully implicit: the solved body displacement feeds straight back.
<code>beamAttachmentKinematics</code> <code>dynamicMeshDict</code>	<code>prescribedByRigidBody</code> / <code>solvedByBlockEigen</code>	Decides whether <code>moorFV</code> or the matrix sets the beam end. Setting both would over-constrain it, which caused an earlier non-convergence.
<code>blockEigenAuthoritativeMotion</code> <code>dynamicMeshDict</code>	<code>false</code> / <code>true</code>	The <code>BlockEigen</code> body solution replaces the 6DOF update; restraint force is zeroed.

With all defaults the code behaves as before. `solvedByBlockEigen` without kinematic coupling stops with a fatal error; kinematic coupling without `BlockEigen` owning the beam end prints a warning and is ignored.

```
// Current stable configuration (tutorials/rigidBodyAndBeam, _fullTime, _shortTime)
blockEigenForceCoupling      true;
blockEigenKinematicCoupling  true;
blockEigenKinematicCouplingMode staged;
beamAttachmentKinematics     solvedByBlockEigen;
blockEigenAuthoritativeMotion true;
```

4. Development timeline

Date	Milestone
Feb 2026	6×6 identity block appended to the matrix; 6DOF input/output vector; basic 6DOF equations and source terms.
10 Apr	Rigid-body coupling support in the beam solver; <code>moorFV</code> passes body state into the beam coupling.
30 Apr – 1 May	First working <code>rigidBodyAndBeam</code> case and benchmark results.
5 – 7 May	<code>BlockEigen</code> body motion logged; shown to match the 6DOF solver with the same forcing; made authoritative (<code>blockEigenAuthoritativeMotion</code>).
24 Jul – 10 Aug	Attachment force history; explicit beam-end ownership (<code>beamAttachmentKinematics</code>); staged mode; fix for divergence at $t = 0.02$ (staged target applied before matrix assembly).
11 Aug	Body→beam coefficient changed to the Newmark-scaled Hooke's-law form; implicit torque rows removed.
19 Aug – 30 Sep	Efficiency plotting script (Newton iterations per step, per interval, cumulative).

5. Validation

Benchmark: `rigidBodyAndBeam_fullTime` (new, staged, BlockEigen authoritative) against `rigidBodyAndBeam_compareSixDoF_Eigen` (old partitioned scheme), single beam, single floating box, serial, $t = 0 \rightarrow 0.25$ s.

Case	Time steps	Newton iterations	Mean / step	Run time
New: monolithic, staged	88	439	4.99	354 s
Old: partitioned	88	439	4.99	351 s

- **Accuracy:** with the same forcing, the BlockEigen body displacement matched the 6DOF solver (checked 7 May; final centre of rotation agreed to ~ 10 significant figures at $t = 0.25$).
- **Cost:** identical. Both cases use `nOuterCorrectors 1`, so the old scheme was already doing a single pass per step. A monolithic solve should pay off where a partitioned scheme needs several passes or goes unstable: light bodies, stiff lines, strong added-mass effects. This benchmark doesn't test that regime.
- **Note:** the `fullTime` run finished at 16:41 on 11 Aug, three minutes before the Hooke's-law commit (16:44). It probably used that code, but this is unconfirmed; a clean rerun would settle it.

6. Limitations and open questions

- **experimental** `fullyImplicit` **mode.** It mixes increments (beam ΔW) and totals (body displacement) in the same equations. Some groundwork is in (`rigidBodyAttachmentDisplacementPrevious_`), but it is not validated.
- **Torque is explicit.** Implicit rotational coupling needs a moment-of-inertia-based Newmark coefficient.
- **Sign convention.** The body $\rightarrow$ beam block signs follow one action/reaction convention. The notes flag this for an audit: if it is wrong, all three signs flip together.
- **Coverage.** Single beam, single body, serial only. Multi-line moorings and parallel runs are untested.
- **Restraint interface.** `restrain(...)` now takes a writable motion object, which lets any restraint modify the body. That should be reviewed before other restraints rely on it.

7. Before merging

	Repo	Item
done	moorFV	Hard-coded <code>/Users/...</code> include paths replaced with <code>\$(BEAMFOAM_DIR)</code> (74005ba).
done	beamFoam	Codex development notes moved from the repo root into <code>codexLogs/</code> .
to do	moorFV	Bring the branch up to date with <code>main</code> (82 commits behind). <code>main</code> renamed <code>sDoFRGBFvBeam</code> → <code>sixDoFRigidBodyBeamMotion</code> , so the restraint files conflict.
to do	moorFV	beamFoam submodule conflict: <code>main</code> points to 4fc1754 (alm-on-main).
to do	moorFV	<code>displacement<name>.dat</code> writes the <i>anchor</i> displacement; rename it or write the attachment displacement.
to do	beamFoam	Remove generated tutorial output (meshes, <code>0/cellLevel</code> , logs, <code>postProcessing</code>): most of the ~127M added lines.
to do	beamFoam	Remove <code>original_BlockEigenSolver0F.C/.H</code> backup and the editor leftover <code>#fixedDisplacementFvPatchVectorField.H#</code> .
to do	both	Put verbose <code>Info</code> diagnostics behind a debug switch; remove commented-out code and the unused <code>PstreamReduceOps.H</code> include.
to do	beamFoam	PR #49 also carries the <code>openfoam-v2306-seevani</code> work (Morison drag, ground contact, spring BC, function objects, cleanup). Merge that first, or accept it bundled.

8. Suggested next steps

1. Clean rerun of `rigidBodyAndBeam_fullTime` against the old case on the current code.
2. A benchmark where partitioned coupling struggles (lighter body or stiffer line, `nOuterCorrectors > 1`), to show whether the monolithic solve helps.
3. Make `fullyImplicit` consistently incremental; add implicit torque rows using the moment of inertia.
4. Audit signs of the body→beam block.
5. Multi-line mooring case (e.g. `mooringCase-H12T20`) and parallel runs.

9. Codex logs in this directory

File	Date	Contents
BLOCK_EIGEN_SOLVER_COUPLING_CHANGES.md	Apr–May	Adding reciprocal body-row coupling terms to <code>BlockEigenSolver0F</code> ; Newmark RHS; moment arm.
BLOCK_EIGEN_RIGID_BODY_COUPLING_PROGRESS.md	7 May	Making the BlockEigen solution authoritative in moorFV; validation against 6DOF.
BEAM_ATTACHMENT_KINEMATICS_OWNERSHIP.md	7 May	Who sets the beam end, and why double ownership caused instability.
rigidBodyKinematicCouplingDevelopment.txt	Jul–Aug	<code>beamAttachmentKinematics</code> switch, config checks, test matrix.
rigidBodyAndBeamCoupledSolverDevelopmentLog.txt	28 Jul	Diagnostics, staged mode, the $t = 0.02$ divergence fix, tutorial cases.
rigidBodyImplicitAttachmentCoefficientChanges.txt	11 Aug	Hooke's-law attachment coefficient; meaning of <code>CQW</code> , <code>pDelta</code> , αK ; sign convention.

These notes were written by Codex during development. Some describe intermediate states that were later superseded (e.g. the implicit torque rows in the first file were removed on 11 Aug). This summary reflects the code at `778e43d` (beamFoam) and `74005ba` (moorFV).